Predicting SNAP Payment Errors: A Pilot Proposal for Texas

Synthetic-data-validated case prioritization for quality-control review

October 4, 2026 · Research whitepaper · Methods and validation artifacts are open-source at github.com/ctrimm/snap-qc-per-experiments

1. Executive summary

We have built and validated a synthetic data generator that reproduces how SNAP payment errors occur in Texas's caseload, calibrated against five years of federal quality-control (QC) microdata. Machine-learning models trained *only* on this synthetic data rank real Texas cases by error risk above chance (binary AUC 0.69 vs 0.67 for a nationally trained model, consistently across seeds) — proof the synthetic cases carry genuine signal about where errors happen. A fully clean refit excluding the holdout rows confirmed the signal is not a contamination artifact (premium: +0.003 on the finding AUC).

The operational finding: a two-part dollar-aware ranking model (error probability × expected dollar impact), applied to prioritize QC review, recovers **$5,798 vs $4,634 for random review** at a fixed review budget — **$67 vs $54 per hour of reviewer time**. Reviewing the riskiest half of cases catches 65% of all errors with half the reviews.

We propose a low-risk pilot: train the ranker on Texas's own historical QC data, validate on a recent holdout, then run it in shadow alongside existing review. No production eligibility-system changes. No new data collection.

2. The problem

SNAP payment error rates remain stubbornly high nationwide, and every error is either taxpayer money over-issued or a household underpaid. Quality-control review is the corrective mechanism — but reviewer time is finite, and cases are currently selected for review without a data-driven sense of where errors concentrate. The result is expensive review effort spread thinly over a caseload where risk is highly uneven.

Texas is a strong candidate for a targeted approach. Our analysis of 3,457 Texas QC cases (FY2020–2024) shows the state's benefit-calculation engine is the cleanest of the three states studied — it matches federal statute at 99.5–100% of holdout cases — so Texas's errors live in the *human process layer* (verification, documentation, coding), which is precisely where a prediction tool adds value. The pattern is distinctive: "other"-coded errors (nature 99) run at 4.28x the national rate and make up nearly a quarter of Texas's errors; agency-caused errors hold the highest share of the three states (58%); and the state's error direction flipped from over-tilted (59% over in FY2020-21) to under-tilted (45–49% over in FY2022-24). Texas's base error rate is the highest of the three states (44%), so even modest ranking gains touch a large volume of cases.

3. What was built and validated

A Texas-conditioned synthetic generator. Starting from a nationally calibrated generator (error-type mix matching reality within ~1 percentage point), we fit Texas-specific error-rate offsets and empirical error mixes. The TX-conditioned output reproduces Texas's finding mix (L1 0.028), element mix (0.034), and cause mix (0.029) — the tightest calibration of the three state tracks — while leaving national output byte-identical.

Validated predictive signal — with one honest qualification. Models trained only on TX-conditioned synthetic cases beat baseline and beat a nationally trained model at ranking error vs correct (AUC 0.688 vs 0.670), and the advantage survived a fully clean refit excluding the holdout rows (0.685 vs 0.670 — a +0.003 contamination premium, effectively zero). On the element-classification task, however, the TX-conditioned model beats the majority baseline (0.407 vs 0.274) but does *not* beat the nationally trained model (0.441): Texas's distinctive signature sits in coding practice (nature-99 "other" coding), which is unlearnable from agency-view features. The transferable signal is real, but it lives in the finding and dollar tasks, not in fine-grained element prediction. Texas-conditioning buys exactly one thing on element: better dependent-care error prediction (F1 0.42 vs 0.38).

A dollar-aware ranking. The set's most important surprise, confirmed a third time: ranking cases by error probability *alone* performs worse than random at the top slice (0.98x). Error probability and error size are nearly orthogonal in Texas's mix — probability chases frequent small-dollar errors (caught median $21) while the dollars live in the income-error tail. The product must therefore be a dollar-aware ranker: here, the two-part structure (P(error) × expected dollars) won — like Pennsylvania, not New York — with lift 1.25x at the top-20% budget.

> **Headline pilot economics (simulated on held-out TX cases, top-20% review budget):**
>
> | | Random review | Dollar ranker |
> |---|---|---|
> | Error dollars recovered | $4,634 | **$5,798** |
> | Value per review hour | $54 | **$67** |
> | Share of all error dollars captured | — | 25.0% |
>
> Of the $5,798: $3,866 is over-issuance (direct program savings); $1,932 is under-issuance (households underpaid — the correct QC outcome, but a make-whole cost). Savings-only value: $44.69/hr vs $32.38/hr random. Honest qualification: the 1.25x lift trails the ~1.5x bar cleared by Pennsylvania (1.58x) and New York (1.52x) — Texas's economics are the weakest of the three tracks, and the whitepaper reports this plainly.

4. The pilot

What Texas provides

- Historical QC case extracts the state already holds (no new data collection, no new PII).
- A review team willing to work a ranked queue for a defined pilot period.

What changes — and what does not

- **Phase 0 (cold start, no TX data needed):** the synthetic-trained ranker scores a sample of cases; rankings are validated against known outcomes.
- **Phase 1:** the dollar ranker is trained on Texas's historical QC data and validated on a recent holdout the model never saw.
- **Phase 2 (shadow):** the ranker scores the live review caseload; reviewers work the top slice; outcomes are measured against the baseline selection method.
- **What does not change:** no modifications to the eligibility system, no automated determinations (the model prioritizes review — it never decides a case), no data leaves the state's control.

Success metrics

- Precision in the reviewed slice vs baseline selection.
- Error dollars recovered per reviewer hour vs baseline.
- Reviewer time saved to reach a fixed error-capture target.

A reasonable pilot shape is 8–12 weeks from data handoff to shadow-readout.

5. Limitations and honest caveats

- **The lift is modest.** The synthetic-trained ranker reaches 1.25x at the top-20% budget — real dollars per hour ($67 vs $54), but below Pennsylvania's 1.58x and New York's 1.52x. The pilot's Phase 1 (training on real TX data) exists to close this gap; the headroom arm projects 1.61x.
- **TX-conditioning does not win everywhere.** On fine-grained element prediction the nationally trained model beats the TX-conditioned one (0.441 vs 0.409); Texas's most distinctive errors (nature-99 "other" coding) are a coding practice, not a learnable feature pattern. The transferable signal lives in finding and dollar tasks.
- **The error direction flipped.** Texas's errors went from 59% over-issuance (FY2020-21) to 45–49% over (FY2022-24). The ranker was validated across this flip (finding accuracy rose +0.086), but the dollar ranker should be re-tuned on recent data at pilot start — under-issuance exposure is the growing half of the problem.
- **A labeling break in FY2024.** Texas went from blanking all nature-97 error causes (FY2020-23) to recording all of them (FY2024). Any cause-mix analysis — and any future generator refit — must stratify by fiscal year.
- **Procedural errors are invisible.** Published federal error rates run above what case microdata implies, in procedural errors the microdata cannot see. This tool targets substantive case errors, not procedural compliance.
- **Stability.** The signal is year-to-year stable in our tests (element accuracy rises +0.034 across the FY2022/23 boundary), but the error-probability baseline should be re-tuned on recent data at pilot start; arithmetic and dependent-care errors are the two buckets with measurable decay and should be watched.
- **Human judgment stays in the loop.** This is a triage tool for reviewers, not an adjudication engine. Final determinations remain human — by design.

6. The ask

We are looking for a Texas partner to run this pilot: a data contact who can provide historical QC extracts, and a review team for the shadow phase. We bring the validated methodology, the synthetic cold-start model, and the open-source validation artifacts. All methods, experiments, and results are published openly; nothing about the approach is proprietary or black-box.

Methods, experiment chronicles (70+), and validation code: github.com/ctrimm/snap-qc-per-experiments (private research repo; access on request). This work is conducted in the open-source civic-technology lane. Contact: via the repository.